

PARAKH SHINDE

Security Engineering | Detection Engineering | Cloud Security | Threat Detection

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Summary

Early-career security engineering candidate who builds and validates detection and cloud-security tooling rather than only completing coursework. Authored and tested 100+ detection rules across Splunk SPL, Sigma, and Elastic KQL mapped to MITRE ATT&CK, and built GCP-IAMGraph, an IAM privilege-escalation analyzer with a three-state authorization engine, eight escalation-detection rules, deterministic attack-graph export, and SARIF/GitHub Code Scanning integration. Background includes 3 years supporting Windows/Linux infrastructure and Active Directory, which informs practical, systems-level detection logic. Currently deepening Python and PowerShell for security automation.

Core Competencies

Detection Engineering: MITRE ATT&CK mapping, SPL/Sigma/KQL rule authoring, detection validation via controlled attack simulation

Cloud Security: IAM privilege-escalation analysis (GCP), CloudTrail/VPC Flow Logs/WAF-based monitoring (AWS)

Threat Detection & IR: log correlation, incident triage, root-cause analysis, NIST SP 800-61 principles

Network Security: TCP/IP, DNS, DHCP, VPN, packet/traffic analysis, IDS signature review

Security Automation: alert normalization, event validation, structured incident-record generation

Technical Skills

Detection & SIEM: Splunk SPL, Microsoft Sentinel (KQL), Sigma, Elastic KQL

Cloud: AWS (CloudTrail, IAM, VPC Flow Logs, WAF), GCP (IAM policy analysis)

Network Security Tools: Wireshark, Sysmon, Suricata, Snort, Zeek

Frameworks: MITRE ATT&CK, NIST CSF, NIST SP 800-61

Programming/Scripting: Python, PowerShell, Bash

Systems: Windows, Linux, Active Directory, Group Policy

Security Projects

GCP-IAMGraph — Explainable IAM Attack-Path Analyzer

Stack: Python, GCP IAM API, SARIF 2.1.0, GitHub Code Scanning

- Problem: Google Cloud IAM effective permissions are difficult to reason about once allow bindings interact with inherited deny policies, hiding exploitable privilege-escalation paths.
- Built a three-state authorization engine that evaluates allow, deny, and inherited-deny policy interactions to compute effective permissions per identity.
- Implemented eight privilege-escalation detection rules targeting known GCP IAM misconfiguration patterns (e.g., impersonation chains, overly broad role bindings).
- Designed deterministic attack-graph export so the same input always produces the same graph, making escalation paths explainable and reproducible rather than a black-box risk score.
- Integrated SARIF 2.1.0 output with GitHub Code Scanning so IAM findings surface directly in a repository's Security tab alongside standard code-scanning alerts.

Enterprise Cloud Security Monitoring & Threat Detection

Stack: AWS CloudTrail, VPC Flow Logs, WAF, IAM, Splunk

- Problem: cloud identity and network activity across CloudTrail, VPC Flow Logs, and WAF were siloed, making IAM abuse and unauthorized API activity hard to investigate centrally.
- Designed and deployed an AWS + Splunk monitoring pipeline ingesting CloudTrail, VPC Flow Logs, WAF, and IAM telemetry into a single investigation surface.
- Engineered 50+ SPL detections for IAM abuse, unauthorized API activity, risky configuration changes, and suspicious access patterns, mapped to MITRE ATT&CK.
- Built dashboards and investigation workflows to support cloud threat hunting and incident response triage.

Enterprise SOC Monitoring & Threat Detection Platform

Stack: Splunk, Sysmon, Suricata, Snort, Zeek, Wireshark

- Problem: needed a realistic, multi-source telemetry environment to develop and validate detection logic rather than relying on vendor sample data.
- Built a production-style SOC lab integrating Windows/Sysmon, Linux, web, IDS, and network telemetry into Splunk for centralized detection and investigation.
- Engineered 50+ SPL detections mapped to MITRE ATT&CK covering reconnaissance, credential abuse, web attacks, privilege escalation, and suspicious PowerShell activity.
- Validated detection logic through controlled attack simulations, cross-referencing Suricata/Snort/ZeeK alerts against Splunk telemetry to reduce false positives.

Security Alert Automation & Incident Triage Workflow

Stack: Splunk webhooks, Python

- Problem: raw Splunk alerts lacked consistent structure, sensitive-field handling, and severity context needed for repeatable triage.
- Built a webhook-driven automation pipeline that normalizes Splunk alert events, validates fields, redacts sensitive data, and assigns severity before generating structured incident records.
- Added analyst-approval gates before any automated response action, prioritizing safe automation over full auto-remediation.

Education

SRM Institute of Science and Technology (SRMIST), Chennai — Master of Computer Applications (MCA),
Cyber Security & Cyber Forensics *Expected Jun 2028, In Progress*

Rabindranath Tagore University (RNTU), Bhopal — Bachelor of Computer Applications (BCA) 2025

Certifications

Completed: Certified Ethical Hacker (CEH v13) Computer Hacking Forensic Investigator (CHFI) Introduction to Offensive Security with Artificial Intelligence — Red Team Leaders)
Advanced Cybersecurity Program — IIT Guwahati (via AccioJob)